



Influence of brazing temperature on microstructural characteristics and interfacial bonding strength in TA15/Ag-Cu-Ti (filler metal) /WC-xCo composites



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Abstract

The development of titanium alloy/cemented carbide bimetal laminated composites aims to integrate the mechanical performance advantages of titanium alloys with the wear-resistant characteristics of cemented carbides. This material system exhibits significant application potential in critical fields such as aerospace, maritime transportation, national defense. Therefore, achieving efficient and stable fabrication of titanium alloy/cemented carbide laminated composites with superior interfacial bonding properties has become a key research focus. This study used Ag-Cu-Ti (filler metal) to prepare TA15/carbide composite plates. The Influences of different brazing temperatures and different Co contents on the interface structure and interface shear

strength were investigated. The results demonstrate distinct interfacial shear strength behaviors for different Co contents. For the low-Co cemented carbide (WC-8Co), the interfacial shear strength initially increases then decreases with rising brazing temperature, peaking at 183.1 MPa at 850°C. In contrast, the high-Co cemented carbide (WC-20Co) exhibits a continuous decrease in interfacial shear strength with increasing temperature, achieving its maximum strength of 238.2 MPa at 800°C. These results offer a scientific basis for the development of high-performance TA15/WC-xCo composites with excellent interfacial integrity via brazing technology.



Keywords

TA15; WC-xCo; Layered material; Brazing

1. Introduction

TA15 titanium alloy, a typical $\alpha+\beta$ type titanium alloy, exhibits outstanding comprehensive mechanical properties, excellent thermal stability, and remarkable fatigue resistance. This alloy plays a critical role in high-end equipment applications, including aero-engine blades, spacecraft load-bearing components, and marine pressure-resistant hulls [1], [2], [3]. This alloy, through a reasonable proportion of elements such as Al, Mo, and V, not only maintains the inherent lightweight advantage of titanium alloys but also significantly enhances the material's strength at room temperature and medium to high temperatures. Its long-term working temperature can exceed 500°C [4,5]. As the demand for multi-functional materials grows, the application of traditional TA15 titanium alloy in specific environments is increasingly limited [6]. For example, the poor wear resistance of titanium alloy surfaces has hindered their practical use in the aerospace, maritime transportation and national defense. [7]. Consequently, researchers are focused on developing gradient functional materials based on TA15. Through techniques such as surface nanoscale treatment [8], laser cladding of ceramic coatings [9], and the formation of layered composite materials with high-strength and high-hardness metals [10], [11], [12], the surface wear resistance of titanium alloys can be significantly improved. This technical approach, which combines “matrix performance optimization and surface function enhancement”, not

only preserves the core advantages of TA15 titanium alloy but also achieves a cross-scale improvement in material performance through multi-scale structural design. This provides key material support for its application and promotion in aerospace, national defense, military industries, and other fields. Notably, the wear-resistant alloy/TA15 bimetallic layered material, created using advanced connection technologies like lamination and compounding, has shown irreplaceable value in specific applications [\[13\]](#), [\[14\]](#), [\[15\]](#).

Through research, cemented carbides have emerged as a key material in industrial applications due to its exceptional wear resistance. This material is composed of high-hardness WC particles as the matrix, which forms a dense structure through metal bonding phases such as Co and Ni. Its Rockwell hardness exceeds HRA90, significantly higher than that of ordinary alloy steel. Under friction and wear conditions, cemented carbide demonstrates an extremely low wear rate. In particularly harsh environments like dry friction and high-speed cutting, its wear resistance is 5–10 times greater than that of tool steel. This outstanding performance is primarily attributed to the high hardness (HV2400) of the WC phase and the excellent toughness of the bonding phase, allowing the material to withstand abrasive cutting without being prone to brittle spalling. Currently, cemented carbide is extensively used in wear-resistant components such as mining drill bits, cutting tools, and molds, offering irreplaceable advantages in enhancing tool lifespan and production efficiency [\[16\]](#), [\[17\]](#), [\[18\]](#).

Traditional plastic processing methods face significant challenges in the preparation of TA15 titanium alloy/tungsten-cobalt cemented carbide composite plates. Due to its poor plastic deformation capacity, large plastic deformation processes such as rolling [\[19\]](#) or co-extrusion [\[20\]](#) are unsuitable for the composite forming performance of cemented carbide. Secondly, there is a significant difference in the coefficients of thermal expansion between TA15 and cemented carbide, which presents a major challenge to achieving a good composite.

To overcome this technical challenge, researchers are focusing on exploring new composite processes, with particular emphasis on addressing the issue of combining dissimilar materials and enhancing interfacial bonding strength.

Currently, there are three main connection methods: the first involves brazing technology using intermediate layer materials [\[21,22\]](#), the second utilizes the laser cladding process based on high-energy beams [\[18,23\]](#), and the third is metallurgical bonding achieved through solid-state diffusion [\[24,25\]](#). For example, Habibi et al. [\[26\]](#) fabricated WC-Co/AISI4140 steel joints using brazing and investigated the impact of brazing parameters, such as temperature and time, on the phase transformation and interfacial microstructure of the welded joints. Their findings indicated that, compared to fusion welding, low-temperature

brazing is a more suitable method, as it prevents the formation of destructive carbides at the metal-ceramic interface. Additionally, Chen et al. [27] applied laser cladding to prepare high-density WC-Co composite coatings on the surface of Ti-6Al-4V, with experimental results showing a strong metallurgical bond between the coatings and the substrate. The coating also exhibited a high microhardness of up to 1536 HV_{0.5}. Furthermore, Shi et al. [28] fabricated WC-Co cemented carbide/tool steel joints with a Ni intermediate layer using diffusion bonding. The results demonstrated that the addition of a Ni intermediate layer effectively suppressed the formation of brittle phases and cracks.

In previous studies, we used the vacuum hot-pressing method to achieve the diffusion bonding of TA15 and WC-xCo [29]. However, when preparing TA15/WC-xCo composite plates using this method, pressure needs to be applied, the preparation efficiency is low, and the cost is high, which hinders its application and promotion. From the research mentioned above, it is evident that Habibi et al. [26] fabricated WC-xCo/steel joints through brazing and achieved a good metallurgical bond. Therefore, with reference to the methods reported in the above-mentioned literature, the TA15/WC-xCo composite plate can be prepared through brazing.

However, there are currently few studies on the preparation of TA15/WC-xCo composite plates via brazing. Ozbaysal et al. [30] found through research that the titanium alloy/WC-Co joint brazed by Ti-Cu-Ni (filler metal) had a low bonding strength due to the formation of a brittle phase interface. Secondly, the Au-Cu-Ni (filler metal) is not suitable for welding titanium alloy/WC-Co joints due to its high cost, poor wettability issues and relatively high melting point. However, Barrena et al. [31] used Ag-Cu-Ti (filler metal) to connect the WC-Co/Ti6Al4V joint. The study found that it could achieve good bonding with both metals, indicating that it has good wettability for WC-Co and Ti6Al4V. In addition, Ong et al. [32] also fabricated the Ti-6Al-4V/Si₃N₄ joint using Ag-Cu-Ti (filler metal) and analyzed the residual stress of the joint in combination with finite element simulation. The results showed that the Ag-Cu phase exhibited better ductility and could alleviate the residual stress between Ti-6Al-4V and Si₃N₄ ceramics. Therefore, the Ag-Cu phase layer, with its excellent plasticity, can act as a plastic buffer layer in the brazing seam, absorbing and dissipating most of the thermal stress (caused by the difference in the coefficient of thermal expansion (CTE) between TA15($\sim 8.9 \times 10^{-6} \text{K}^{-1}$) and WC-xCo ($5.5\text{--}6.5 \times 10^{-6} \text{K}^{-1}$)) through its own plastic deformation, thereby significantly reducing the tendency to crack and improving the reliability of the joint. Secondly, in this study, TA15 titanium alloy was used as the base material, with a phase transformation point of approximately 1000°C. Ag-Cu-Ti (filler metal), with a melting point of around 780°C, was selected. After brazing, the properties of the TA15 titanium alloy base material were well maintained. Therefore, for the research questions, this study focuses on the preparation of TA15/WC-xCo composite plates using Ag-Cu-Ti

(filler metal) and explores the effects of various brazing temperatures and Co content on the interface structure and bonding strength of the composite plates.

2. Experimental materials and methods

2.1. Materials

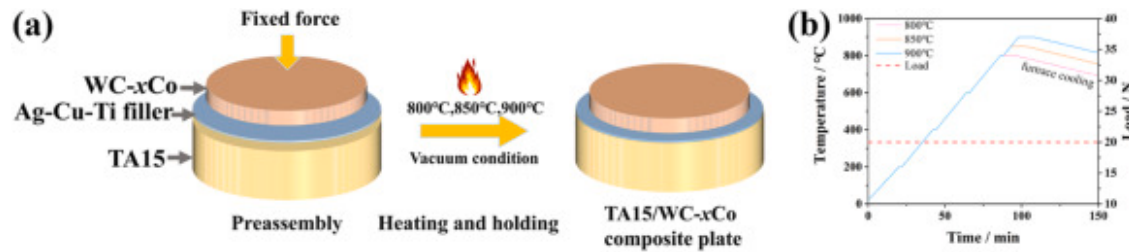
The test used forged TA15 plates ($R=19.5\text{ mm}$) with a thickness of 16 mm, cemented carbide ($R=15\text{ mm}$) with low Co content (WC-8Co) and high Co content (WC-20Co) with a thickness of 2 mm (WC-xCo), and Ag-Cu-Ti (filler metal) with a thickness of 0.05 mm. The specific compositions are provided in [Table 1](#).

Table 1. Chemical Composition and Content (wt%) of TA15, WC-xCo, and Ag-Cu-Ti (filler metal).

Materials	Element						
	Co			WC			
WC-8Co	8.4			bal.			
WC-20Co	20.3			bal.			
TA15	Element						
	Al	Mo	V	Zr	Fe	C	Ti
	6.8	1.8	2.2	2.3	0.05	0.01	bal.
Ag-Cu-Ti	Ag			Cu		Ti	
	68.85			bal.		4.73	

2.2. Brazing process

The brazing process was designed based on the melting point of the brazing filler metal. The preparation process and process parameter curves are shown in Fig. 1. The influence of two types of cemented carbides, with low Co content (WC-8Co) and high Co content (WC-20Co), as well as different brazing temperatures (800°C, 850°C, 900°C), on the interface structure and bonding performance is discussed. During the preparation of the composite plate, the surfaces of the TA15 alloy, cemented carbide, and Ag-Cu-Ti (filler metal) were first ground to remove surface stains and oxide layers. The ground samples were then immersed in an alcohol solution and subjected to ultrasonic cleaning to ensure the material surfaces were clean. Subsequently, the materials were stacked in the order shown in Fig. 1(a), and a relatively small fixing force (10–30N) was applied to the top of the sample. Heating and holding were then carried out under vacuum conditions (1E^{-2}) to complete the preparation of the TA15/WC-xCo composite plates. The temperature changes during the brazing process are shown in Fig. 1(b). After heating at a rate of $10^\circ\text{C}/\text{min}$ to the preset temperature (800°C, 850°C, or 900°C), the samples were kept for 10min and then cooled to room temperature (furnace cooling) for approximately 7–8h.



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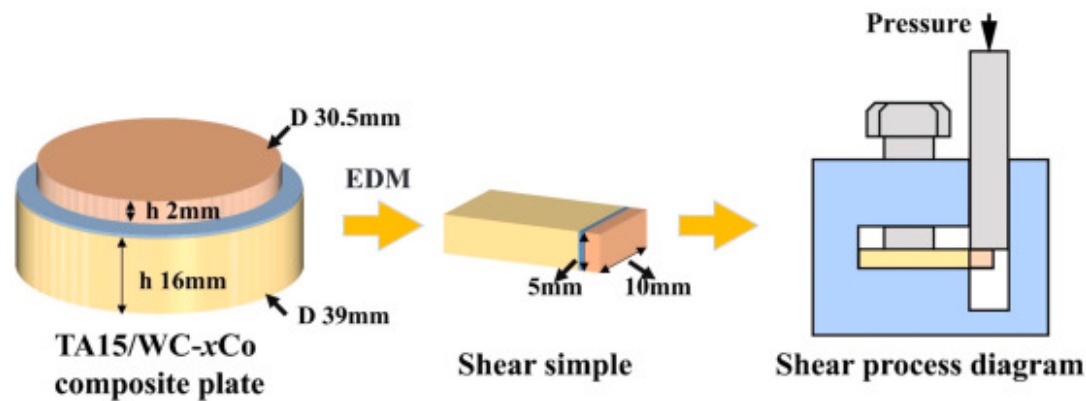
Fig. 1. The schematic diagram of the process for preparing TA15/WC-xCo composite plates based on brazing and a graph of the process parameters, (a) Preparation process, (b) process parameter curve.

2.3. Performance testing and tissue characterization

On the one hand, the loss rate of the matrix performance after brazing of TA15/WC-xCo composite plates is an important criterion for evaluating the preparation process. Therefore, a universal testing machine was used to conduct tensile tests on TA15 titanium alloy under different conditions to assess its post-weld strength and elongation loss rate. Meanwhile, crystallographic analysis of the microstructure of TA15 titanium alloy at different brazing temperatures was conducted using backscattered electron diffraction (EBSD).

On the other hand, the stability of the interface structure and the bonding strength of TA15/WC-xCo composite panels are key factors in evaluating their quality. Firstly, to analyze the variation of the interface structure of the composite plate with respect to the brazing temperature, scanning electron microscopy (SEM) and energy dispersive spectroscopy (EDS) were used to examine the interface structure under various conditions. Secondly, the reaction enthalpy generated by the interface phase was calculated using the *Miedema* model to aid in analyzing the types of interface reaction phases. Additionally, the performance characteristics associated with different interface structures need to be discussed. Prepare compression shear specimens, conduct shear tests, and evaluate the interfacial bonding performance. The size of the shear test sample (5mm×10mm) and the schematic diagram of the test process are shown in Fig. 2. Prior to testing, the surface of the sample was ground to achieve a smooth finish. During the test, a downward pressure rate of 0.5mm/min was applied. The test was completed when the interface was fully separated. Finally, the interfacial bonding strength was calculated using the shear strength formula, as follows:

$$\tau = F/S$$



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Fig. 2. TA15/WC-xCo composite plate shear sample and shear process diagram.

In the formula, τ represents the interfacial shear strength (MPa). F is the test breaking load (N). S represents the shear surface area of the sample (mm^2).

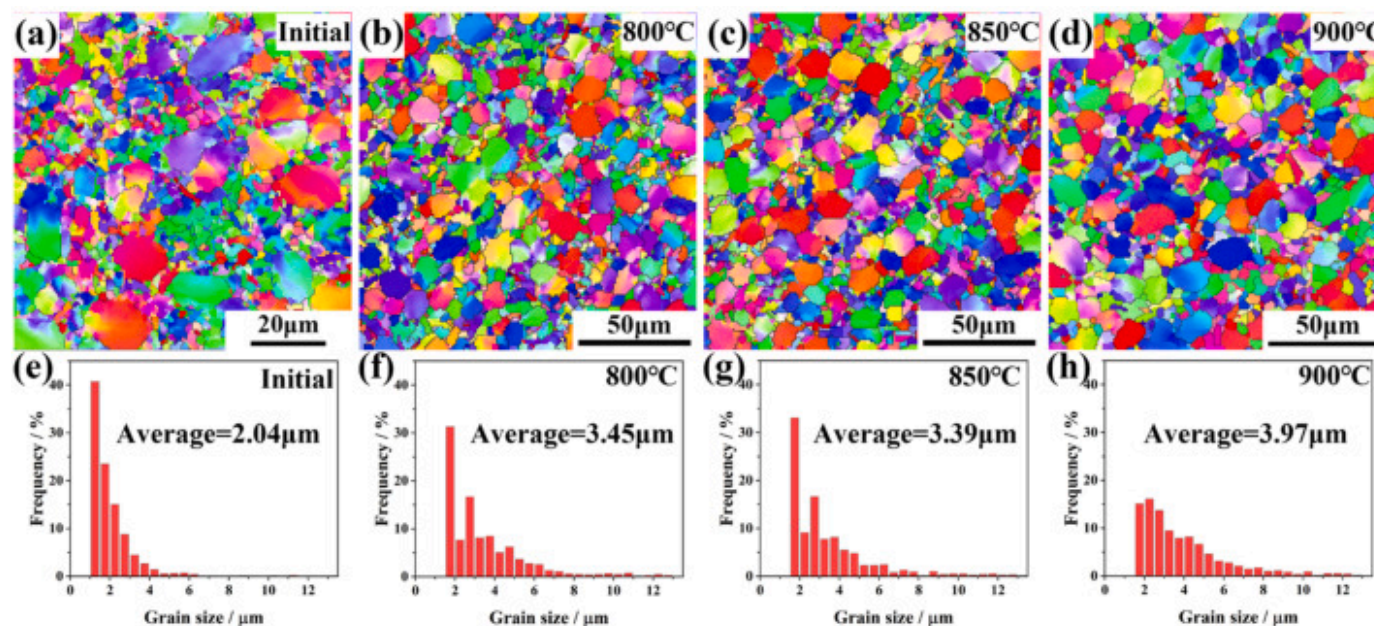
After the test, SEM, EDS and X-ray diffraction (XRD) analyses were conducted on the shear surface. Combined with the above-mentioned interface microstructure, the effects of different brazing temperatures and different Co contents on the interface bonding performance and shear failure were obtained.

Finally, the thermal stress distribution at the interface of the TA15/WC-xCo composite plate after brazing and cooling was simulated by using the finite element simulation method.

3. Results and discussions

3.1. Component material analysis

After brazing, the material properties of WC-xCo do not change significantly. However, for TA15 titanium alloy, its performance is affected by the high-temperature holding and slow cooling process during brazing. Therefore, it is crucial to study the matrix structure and mechanical properties of TA15 after brazing. The microstructure and grain size distribution of TA15 titanium alloy, based on EBSD, are shown in Fig. 3. The original structure consists of a small number of large grains (about 10–20 μm) and a large number of fine grains (1–2 μm), resulting in an average grain size of approximately 2 μm . Additionally, the uneven color within the large grains indicates that the original TA15 structure exhibits different local crystal orientations within the grains due to plastic deformation (forging). After brazing at 800°C–850°C, the grain size of TA15 became relatively uniform, with a reduction in the proportion of 1–2 μm grains, raising the average grain size to approximately 3.4 μm . After brazing at 900°C, the grains of TA15 grew significantly, with an average grain size of approximately 4 μm , mainly consisting of equiaxed grains ranging from 2 to 6 μm . Meanwhile, compared with the initial state, the orientation differences within the large grains are significantly reduced.

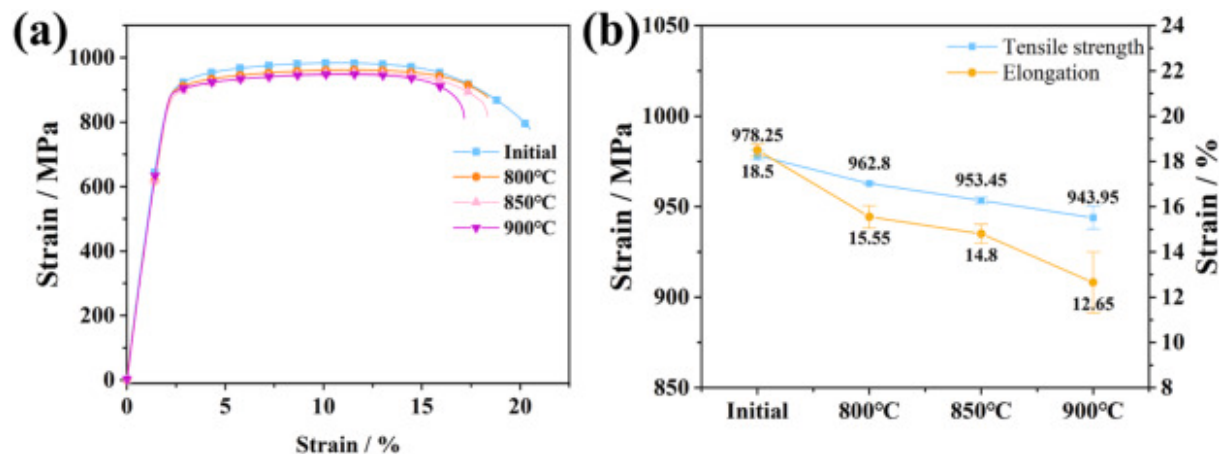


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Fig. 3. Microstructure and grain size distribution of TA15 titanium alloy based on EBSD: (a,e) Initial, (b,f) 800°C, (c,g) 850°C, (d,h) 900°C.

As shown in Fig. 4, the stress-strain curve of TA15 and the variations in tensile strength and elongation with brazing temperature are presented. The average tensile strength and elongation of the original TA15 (calculated from the elongation of the gauge length section after fracture) can reach 978.25 MPa and 18.5%, respectively. After brazing, both the tensile strength (UTS) and elongation of TA15 decreased. Moreover, as the brazing temperature increased, the UTS and elongation gradually declined. The performance decline is also related to the growth of TA15 grains. As the brazing temperature increases, the grain size becomes larger, and the strength and elongation of TA15 decrease accordingly. As the brazing temperature rises, the grains gradually increase in size. So, at 900°C, the excessive grain size significantly affects the UTS and elongation of TA15. Therefore, to better preserve the mechanical properties of the TA15 substrate, the brazing temperature should not be too high.



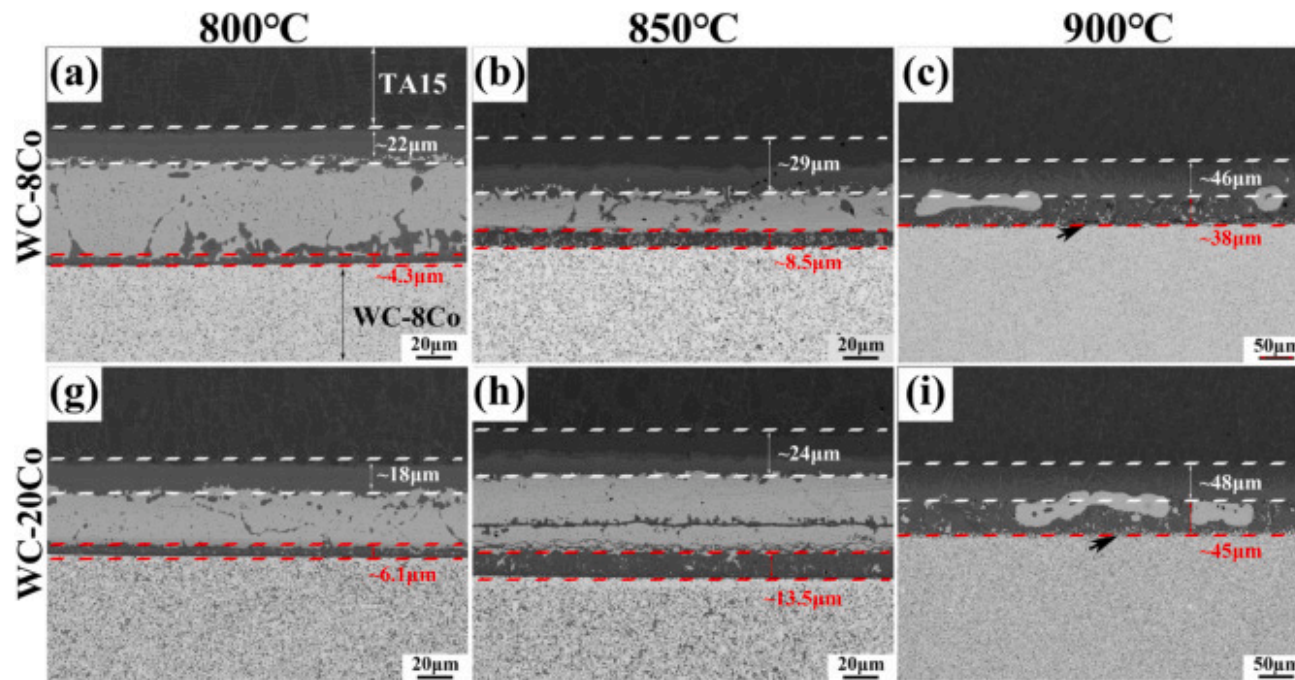
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Fig. 4. Results of engineering stress-strain curve, tensile strength and elongation of TA15 titanium alloy with temperature: (a) engineering stress-strain curve, (b) tensile strength and elongation with temperature.

3.2. Interfacial microstructure and bonding properties

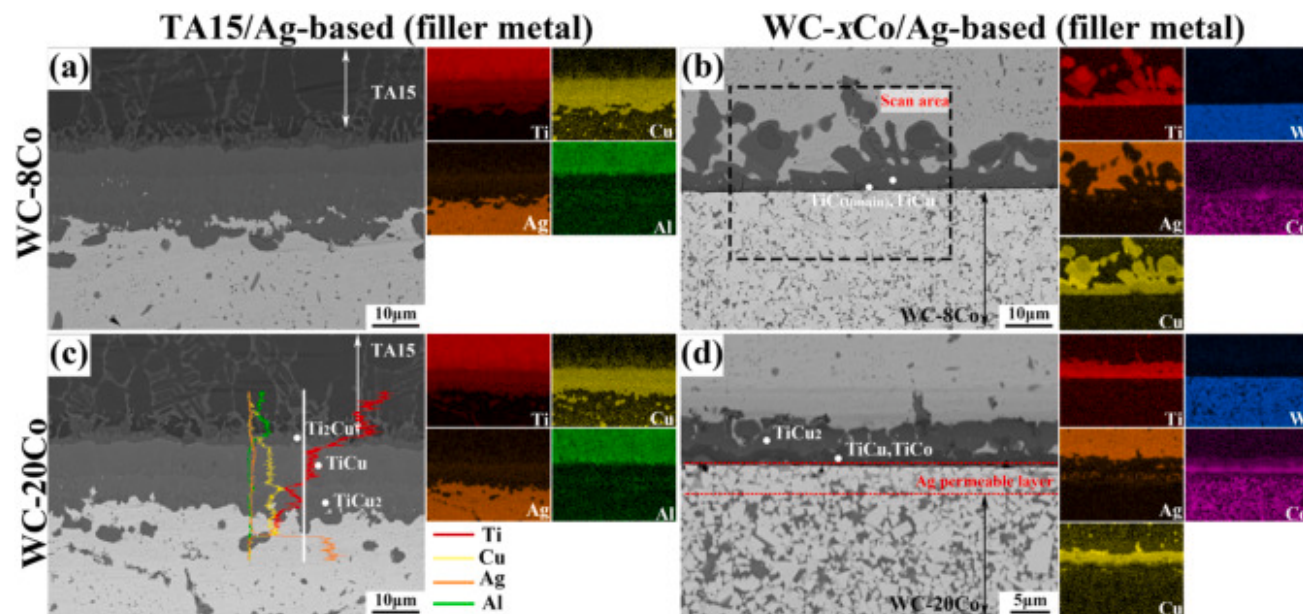
The different brazing temperatures and Co content in WC-xCo have a direct relationship with the interfacial reactions and phase formation. Therefore, the influence of different process parameters and material types on the interfacial microstructure will be analyzed next. Fig. 5 shows the interface layer structure of brazed TA15/WC-xCo composite plates at different temperatures (800°C, 850°C, and 900°C). The interface microstructure of TA15/WC-xCo composite plates brazed at these temperatures was magnified and observed, and energy spectrum analysis was conducted. The results are shown in Fig. 6, Fig. 7, Fig. 8.



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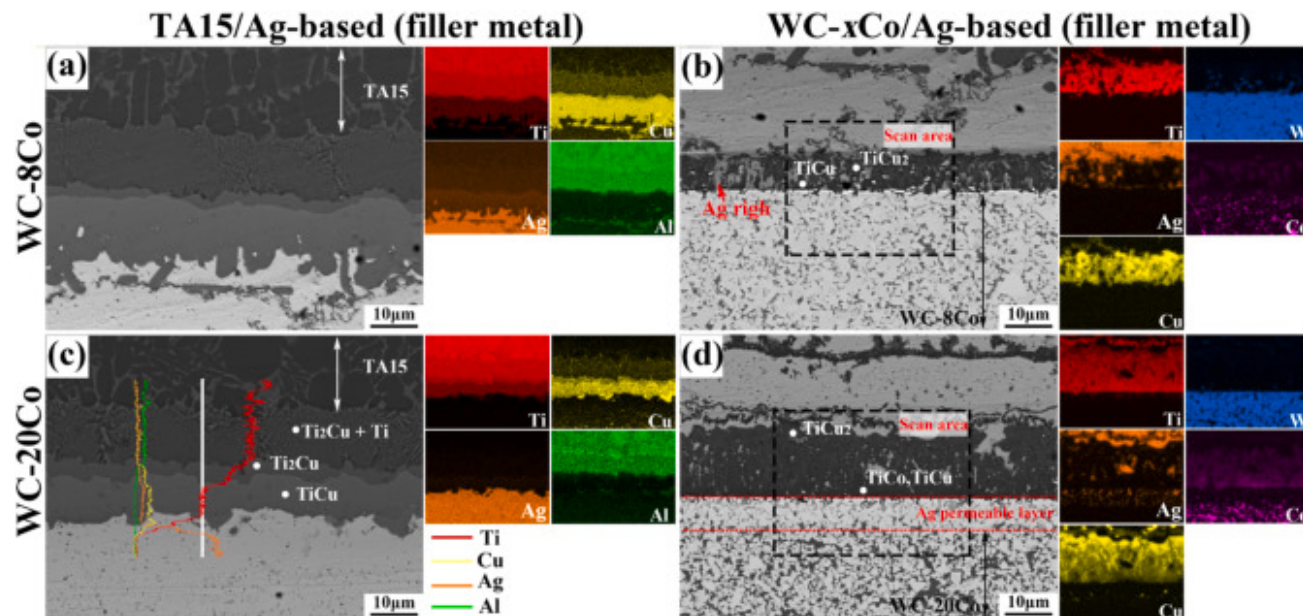
Fig. 5. Braze TA15/WC-xCo composite plate interface microstructure at different temperatures, (a–c) TA15/WC-8Co composite plates brazed at temperatures of 800°C, 850°C and 900°C respectively, (g–i) TA15/WC-20Co composite plates brazed at temperatures of 800°C, 850°C and 900°C respectively.



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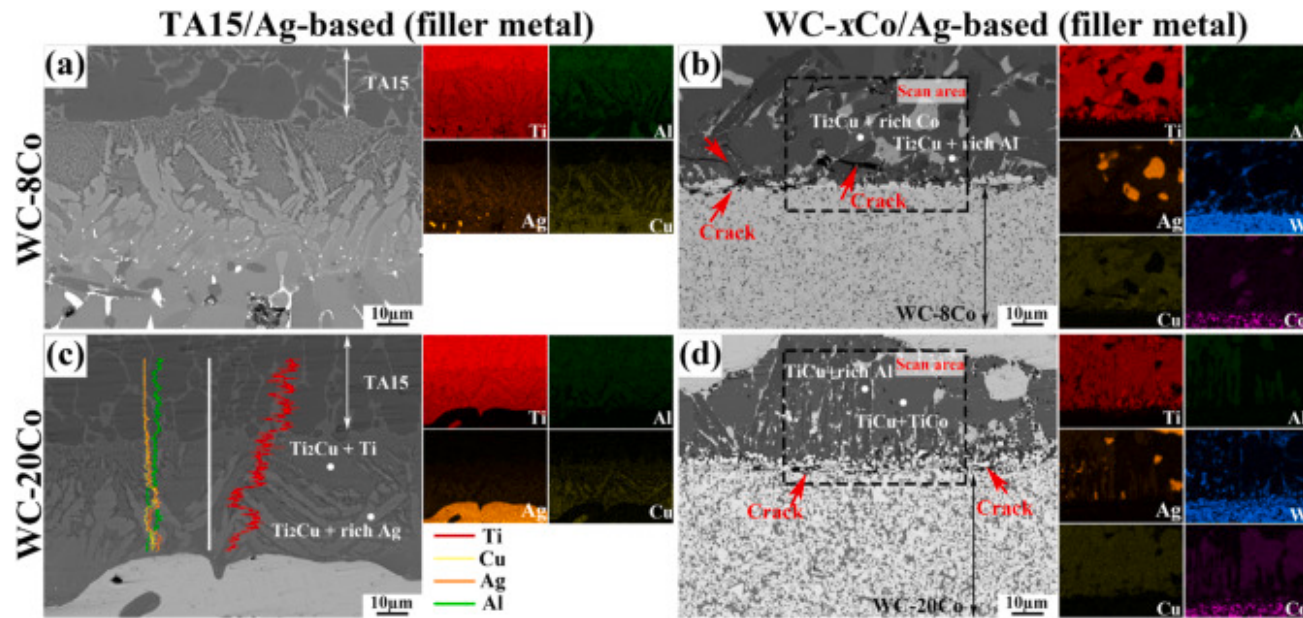
Fig. 6. Interfacial microstructure and EDS results of TA15/WC-xCo composite plates brazed at 800°C, (a) TA15/Ag-based (filler metal) interface of TA15/WC-8Co composites, (b) WC-8Co/Ag-based (filler metal) interface of TA15/WC-8Co composites, (c) TA15/Ag-based (filler metal) interface of TA15/WC-20Co composites (d) WC-20Co/Ag-based (filler metal) interface of TA15/WC-20Co composites.



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Fig. 7. Interfacial microstructure and EDS results of TA15/WC-xCo composite plates brazed at 850°C, (a) TA15/Ag-based (filler metal) interface of TA15/WC-8Co composites, (b) WC-8Co/Ag-based (filler metal) interface of TA15/WC-8Co composites, (c) TA15/Ag-based (filler metal) interface of TA15/WC-20Co composites (d) WC-20Co/Ag-based (filler metal) interface of TA15/WC-20Co composites.



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Fig. 8. Interfacial microstructure and EDS results of TA15/WC-xCo composite plates brazed at 900°C (a) TA15/Ag-based (filler metal) interface of TA15/WC-8Co composites, (b) WC-8Co/Ag-based (filler metal) interface of TA15/WC-8Co composites, (c) TA15/Ag-based (filler metal) interface of TA15/WC-20Co composites (d) WC-20Co/Ag-based (filler metal) interface of TA15/WC-20Co composites.

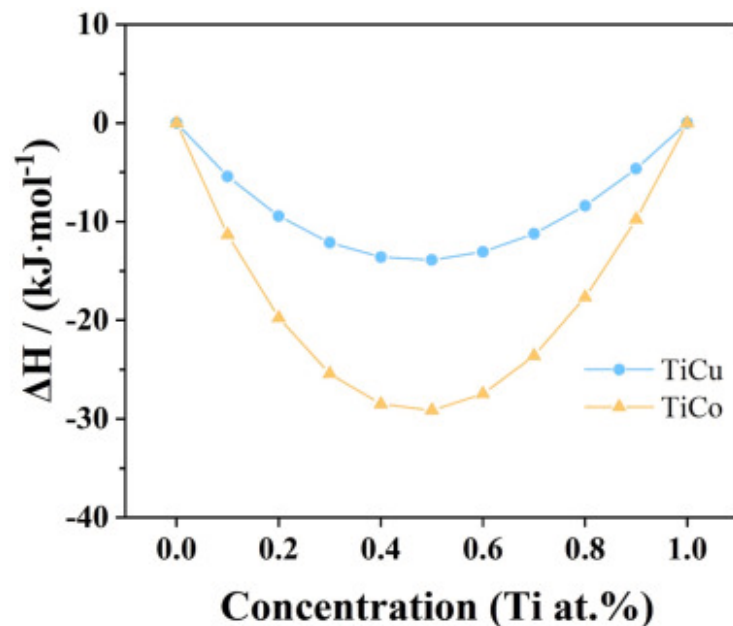
From the perspective of the overall interface structure, at temperatures between 800°C and 850°C, the Ag-based (filler metal) at the composite interface is continuously distributed, whereas at 900°C, the Ag-based (filler metal) at the interface is discontinuously distributed. It might be due to the fact that the Ag-based (filler metal) volatilizes at 900°C, causing its discontinuous distribution. For the TA15/Ag-based (filler metal) interface, the TA15/WC-xCo composite plate exhibits good bonding across different brazing temperatures. As the temperature increases, the thickness of the diffusion layer between the interfaces increases significantly.

The TA15/Ag-based (filler metal) interface under different conditions was magnified for observation, and energy spectrum analysis was conducted. It was found that, at the same temperature, the types of phases formed at the TA15/Ag-based (filler

metal) interface were not related to changes in Co content. At 800°C, in the TA15/WC-xCo composite plate, from the TA15 layer to the Ag-based (filler metal) layer, as the Ti content gradually decreases, the Ti_2Cu ($\sim 2\mu\text{m}$), TiCu ($\sim 8\mu\text{m}$), and TiCu_2 ($\sim 8\mu\text{m}$) layers are successively formed, as shown in Fig. 6. This phenomenon is similar to the results in the literature [33]. Similarly, at 850°C, from the TA15 layer to the Ag-based (filler metal) layer, as the Ti content gradually decreases, the $\text{Ti}_2\text{Cu}+\text{Ti}$ mixed layer ($\sim 13\mu\text{m}$) [33], Ti_2Cu ($\sim 3\mu\text{m}$), and TiCu ($\sim 8\mu\text{m}$) layers are formed successively [34]. However, at 900°C, the morphology of the intermediate phase is uneven, primarily consisting of dendritic Ti_2Cu as the main component, with a small amount of Ag also being enriched in the dendritic Ti_2Cu phase. This might also be a reason for the discontinuous distribution of the Ag-based (filler metal) in the interface.

For the Ag-based (filler metal)/WC-xCo interface, at 800°C–850°C, the interface is tightly bonded without obvious defects. However, at 900°C, noticeable pores appear at the interface, as shown in Fig. 5. Similar to the TA15/Ag-based (filler metal) interface, the thickness of the diffusion layer increases as the temperature rises. Secondly, with the increase of Co content, the thickness of the continuous Ti-Cu and Ti-Co IMCs layers increases accordingly.

Further observation of the interface and energy spectrum analysis was conducted. At 800°C, the Ti element passes through the Ag-based (filler metal) and reacts with Cu and Co elements to form Ti-Cu and Ti-Co intermetallic compounds (IMCs). IMCs such as TiCu_2 , TiCu , and TiCo were formed at the Ag-based (filler metal)/WC-xCo interface [35]. The enthalpy of reaction formation for Ti-Co and Ti-Cu IMCs was calculated based on the *Miedema* model, and the results are shown in Fig. 9. The results indicate that during the brazing process, Ti-Co IMCs form preferentially over Ti-Cu IMCs. Therefore, for WC-20Co, more Co elements diffuse, causing Ag-based (filler metal)/WC-20Co to form more Ti-Co IMCs. It makes the IMCs layer of the Ag-based (filler metal)/WC-20Co interface thicker. Additionally, during the diffusion and loss of Co elements, Ag elements simultaneously diffuse into the WC-20Co matrix, replacing the bonding role of Co. This ensures the integrity of the matrix, and improves the bonding properties of the interface.



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Fig. 9. The enthalpy of interfacial phase reaction formation calculated based on *Miedema* model.

At 850°C, the interface structure of Ag-based (filler metal)/WC-8Co is primarily composed of Ti-Cu IMCs and Ti-Co IMCs, with fine blocky Ag-based (filler metal) and WC fragments diffusely distributed. The interface composition of Ag-based (filler metal)/WC-20Co is similar, but there are no WC fragments in the IMCs layer, and its thickness is significantly increased. According to energy spectrum analysis, the Ti-Cu and Ti-Co IMCs at the interface are mainly composed of TiCu₂, TiCu and TiCo phases. Additionally, the "substitution" behavior of Ag and Co elements is consistent with that observed at 800°C, ultimately achieving a good interface combination and the structural integrity of WC-xCo.

At 900°C, due to the discontinuity of the Ag-based (filler metal), the IMCs layers on both sides of the interface come into direct contact. The analysis of the IMCs layer on the WC-xCo side reveals that it mainly consists of massive Ti-Cu and Ti-Co IMCs as well as WC fragments. However, the size of the IMCs is larger, and obvious crack defects are present in the diffusion layer at the same time. For WC-8Co, based on the analysis of energy spectrum and the XRD results in the following text, it indicates that the IMCs

layer is primarily Ti_2Cu , but different regions are enriched with various elements, such as Al or Co. At the same time, obvious cracks can be observed between the IMCs layers, likely formed by thermal stress during the cooling process. For WC-20Co, the results shows that the IMCs are primarily Ti-Cu, Ti-Co. Meanwhile, the formation of Ti-Co is always accompanied by the generation of Ti-Cu, as shown in Fig. 8(d). Based on the analysis of XRD results in the following text, the formed Ti-Cu and Ti-Co IMCs may include TiCu, TiCo. It is consistent with the literature reports [35,36]. Meanwhile, similar to WC-8Co, TiCo is formed in the TiCu of some regions, and Al elements are enriched in some areas [37]. Secondly, it can be found that WC-20Co near the interface has formed long cracks parallel to the interface. The possible cause of its formation might be the diffusion of Co elements, the reduction of WC-20Co strength near the interface, and the formation of parallel interface thermal cracks during the cooling process.

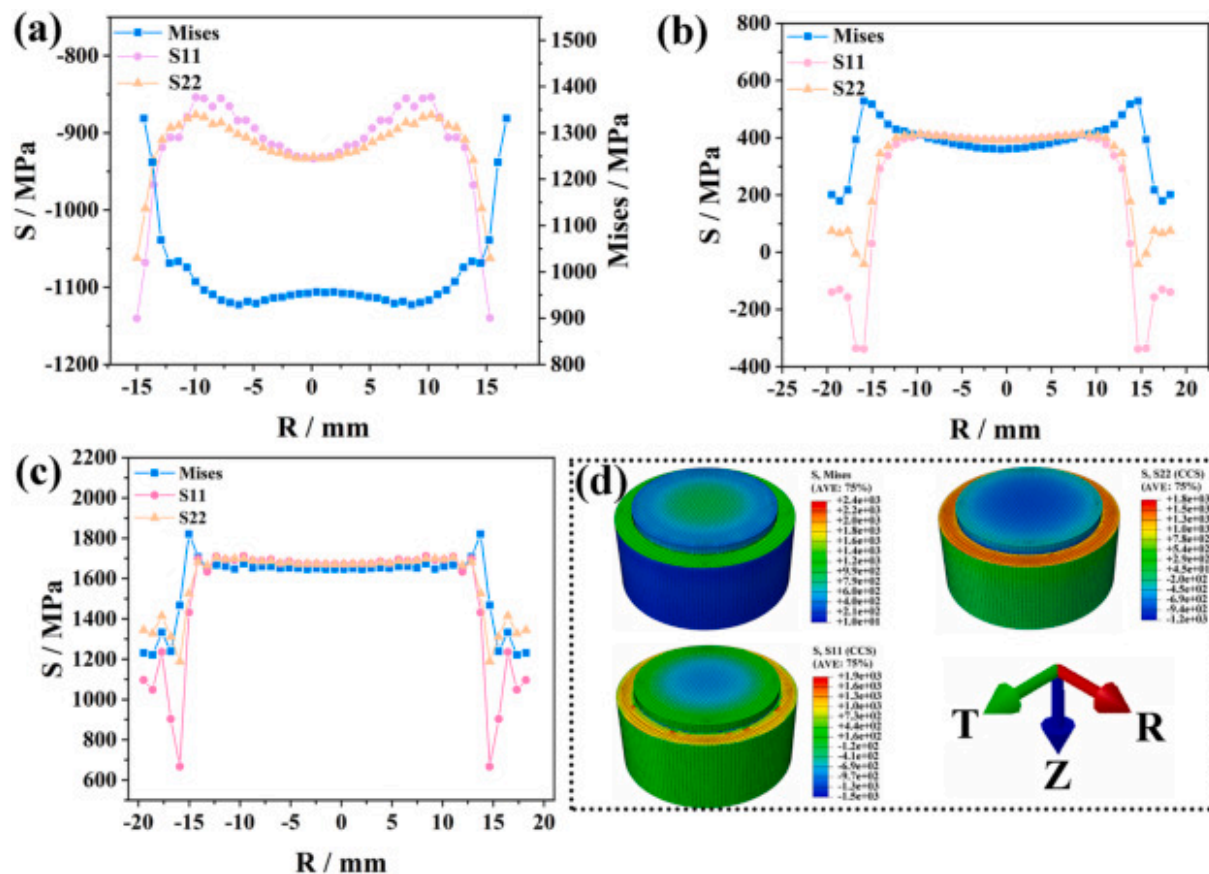
To summarize, at 800°C – 850°C , more Co elements in WC-20Co diffuse to form more TiCo phase, making the IMCs layer at the Ag-based (filler metal)/WC-20Co interface thicker. However, when the temperature rises further to 900°C , the diffusion of more Co elements in WC-20Co reduces the strength of WC-20Co near the interface, forming long cracks, which seriously affects the interfacial bonding performance of the composite plate.

In the above interfacial microstructure analysis, it was found that at 900°C , thermal cracks occurred in both the IMCs layer and the WC-xCo layer of the Ag-based (filler metal)/WC-xCo interface. Therefore, the formation and distribution of interfacial stress were analyzed using the finite element simulation method. Taking the TA15/WC-20Co composite plate brazed at 800°C as an example, the simulation calculation was carried out based on the dimensions of each component material in section 2.1 (The material properties are shown in Table 2. The material properties of TA15 were calculated using *JmatPro*, while other material properties were obtained through literature reports). A total of 69,048 meshes were divided, and the final finite element simulation results are shown in Fig. 10. The stress (Mises, S11, S22) on the diameter of the WC-20Co and Ag-based (filler metal) and TA15 interface was analyzed, and the results are shown in Fig. 10(a and b). Due to the significant difference in the thermal expansion coefficients of TA15 and WC-20Co during the cooling process. The radial shrinkage rate of TA15 and Ag-based (filler metal) towards the center of the sheet is more pronounced, causing WC-20Co to experience radial stress (S11) contracting towards the center, while TA15 and Ag-based (filler metal) undergo radial stress (S11) extending outward from the center. This also applies to the analysis of circumferential stress (S22). Meanwhile, the stress values of Mises at the center of the composite plate are all smaller than those at the edges. It indicates that the central part of the board is less affected by stress. During the cooling process after brazing at 900°C , the interface undergoes greater shrinkage stress. Additionally, due to the diffusion and loss of Co elements in WC-xCo. For WC-8Co, due to its low Co content, the diffusion and loss of Co will not cause the collapse of

the WC "skeleton." During cooling, crack defects mainly form within the IMC layer. For the WC-20Co, near the interface generates a mass of cavity defects, to the extent that long cracks parallel to the interface occur during the cooling process. These have seriously affected the bonding performance of the interface.

Table 2. Relevant Parameters for finite element Simulation Calculation [[38], [39], [40], [41]].

Materials	Temperature (°C)	Elastic modulus (GPa)	CET ($10^{-6} \cdot K^{-1}$)	Poisson's ratio	Density ($g \cdot cm^{-3}$)	Conductivity (W/(m·K))	Specific Heat (J/(g·K))
WC-20Co	20	495	6.5	0.22	13.5	142	0.2
	300					123	0.232
	600					90.4	0.287
	900					75.3	0.321
TA15	20	118	8.9	0.39	4.45	8.0	0.52
	200	108	9.2			10.2	0.587
	400	97	10			12.2	0.67
	600	85	10.9			15.1	0.755
	800	75	10.9			18.0	0.88
Ag-Cu-Ti	20	111	16.7	0.363	9.76	89.7	0.289
	200	90	17.2				
	400	80	17.8				
	600	67	19.4				
	800	58	20.5				



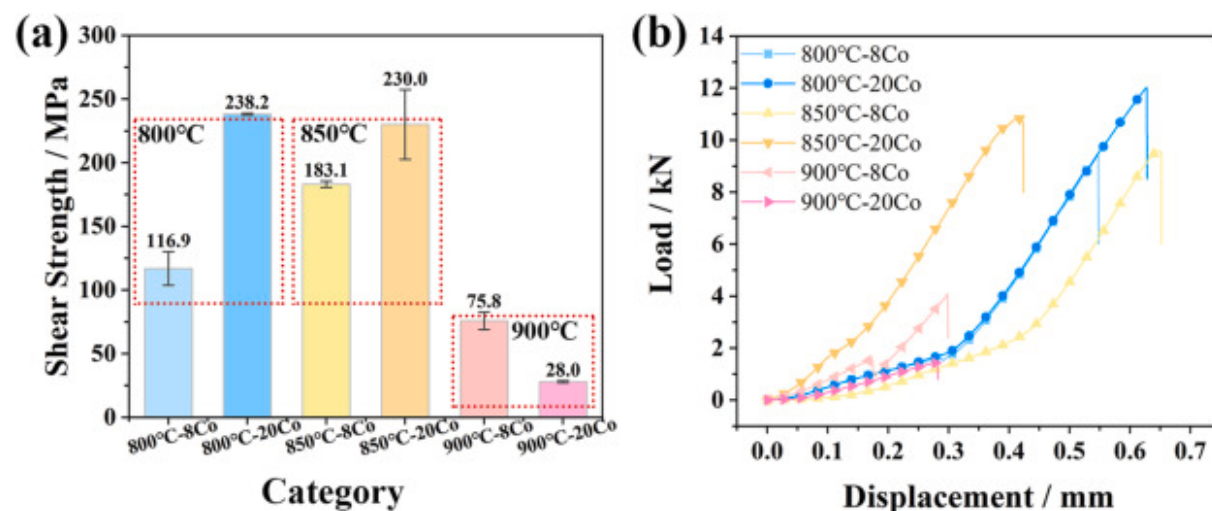
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Fig. 10. Results of thermal stress at the composite interface after brazing based on finite element simulation, (a–c) Stress variation in the diameter direction of WC-20Co, TA15 and Ag-Cu-Ti (filler metal) respectively (d) Stress cloud diagram of TA15/WC-20Co composite plate.

The above analysis has focused on the interface structure and phase composition, but it is also crucial to study the relationship between microstructure and performance. We tested and analyzed the shear performance of composite plates under different brazing conditions and examined their shear failure interfaces.

Fig. 11 presents the test results of the interface shear performance of TA15/WC-xCo composite plates brazed under different conditions. For the TA15/WC-8Co composite plate, as the temperature increases (from 800°C to 900°C), the interfacial shear strength first increases and then decreases, reaching its maximum at 850°C, with a shear strength of 183.1 MPa. For the TA15/WC-20Co composite plate, the interfacial shear strength gradually decreases with the increase in temperature, reaching its maximum at 800°C, with a shear strength of 238.2 MPa. The interfacial shear strength of the composite plates brazed at both 800°C and 850°C does not differ significantly. However, as the temperature increases further, the interfacial shear strength decreases substantially. This is related to the noticeable hot cracks at the interface of the TA15/WC-20Co composite plate brazed at 900°C, as observed earlier.



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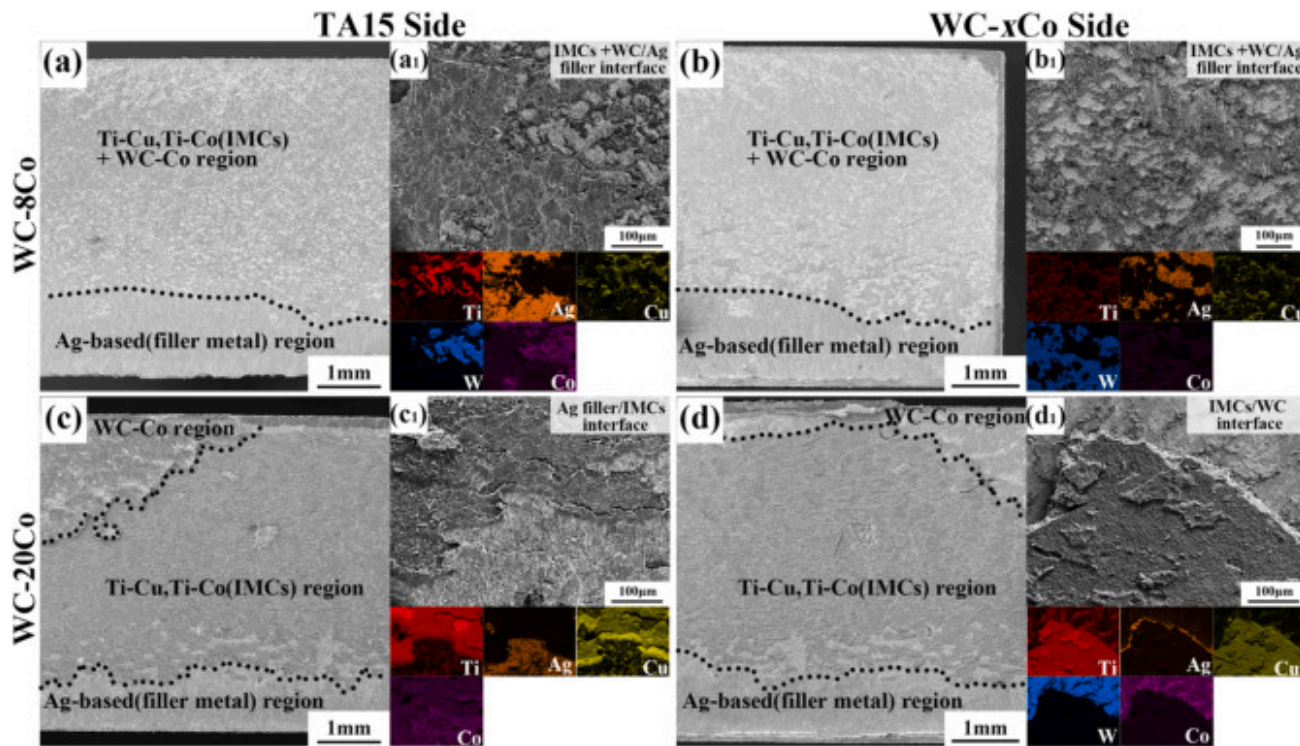
Fig. 11. Interface shear test results of brazed TA15/WC-xCo composite plates at different temperatures, (a) Shear strength, (b) displacement-load curve.

Secondly, at 800°C and 850°C, the interfacial shear strength increases with the rise in Co content. Based on the above analysis of the interface structure, for high-Co content cemented carbide, Co is diffused and lost, while Ag is diffused into WC-20Co, replacing the role of the Co binder. This ensures the integrity of the WC skeleton at the interface while facilitating the mutual diffusion of elements, thereby enhancing the shear strength of the interface. However, at 900°C, as the Co content increases, the

interfacial shear strength decreases. The main reason is that a large amount of Co has diffused and been lost, leaving no binder to stabilize the WC skeleton. During the cooling process, the interface experiences contraction stress toward the center, eventually leading to the formation of long cracks parallel to the interface in the WC-20Co matrix near the interface, resulting in a decrease in the bonding strength.

3.3. Failure analysis

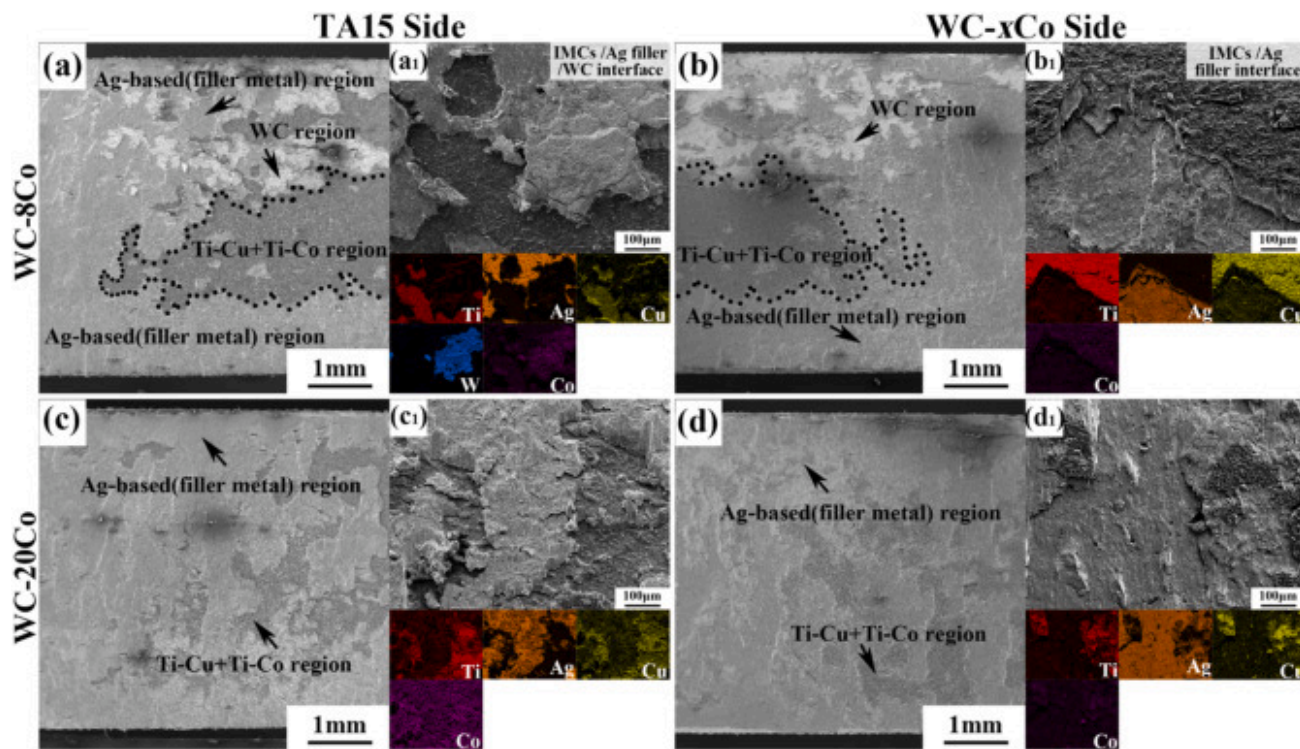
The morphology of the shear surface of the composite plate after the shear test was observed, and EDS analysis was conducted. The results are shown in Fig. 12, Fig. 13, Fig. 14. To assist in analyzing the causes of interfacial shear failure, X-ray diffraction analysis was used to detect the phase composition of the shear surface, and the results are shown in Fig. 15.



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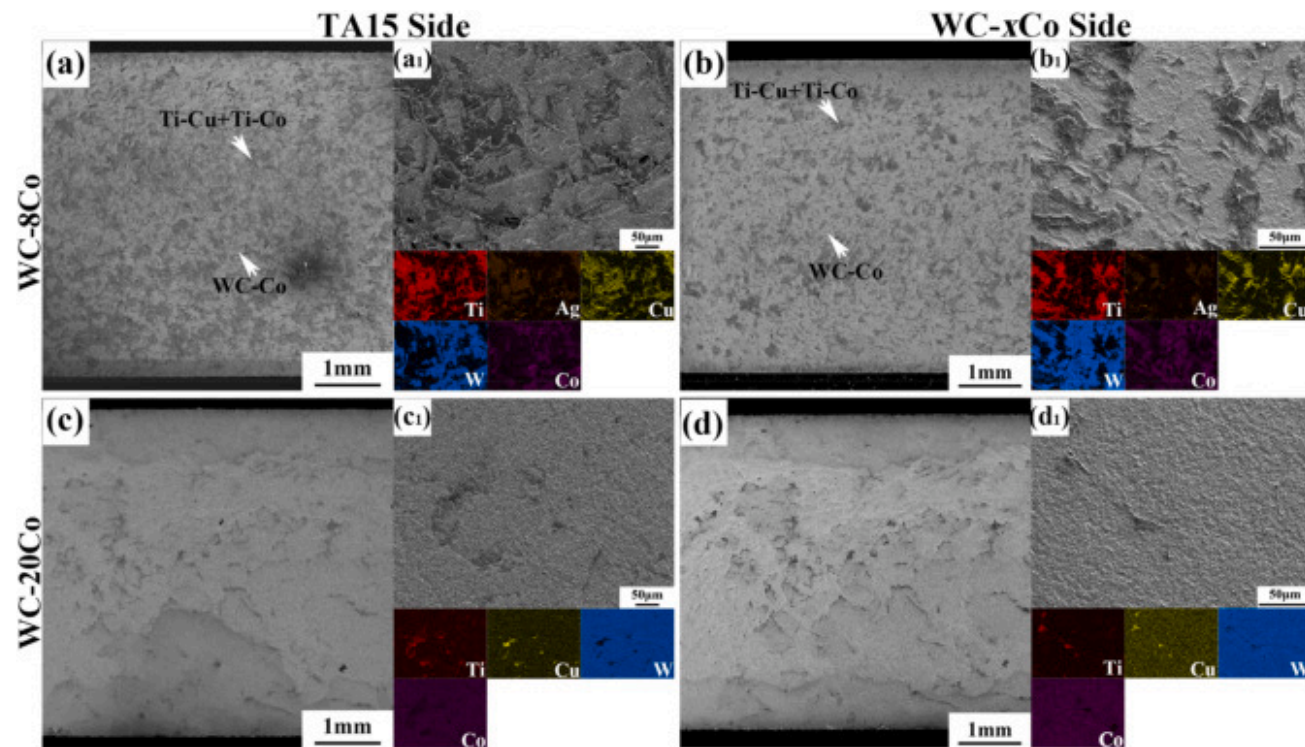
Fig. 12. Shear surface micromorphology and EDS analysis results of TA15/WC-xCo composite plate brazed at 800°C, (a) and (b) are respectively the TA15 side and the WC-8Co side of the TA15/WC-8Co composite plate, (c) and (d) are respectively the TA15 side and the WC-20Co side of the TA15/WC-20Co composite plate, (a₁, b₁, c₁, d₁) are respectively the magnified views of (a, b, c, d).



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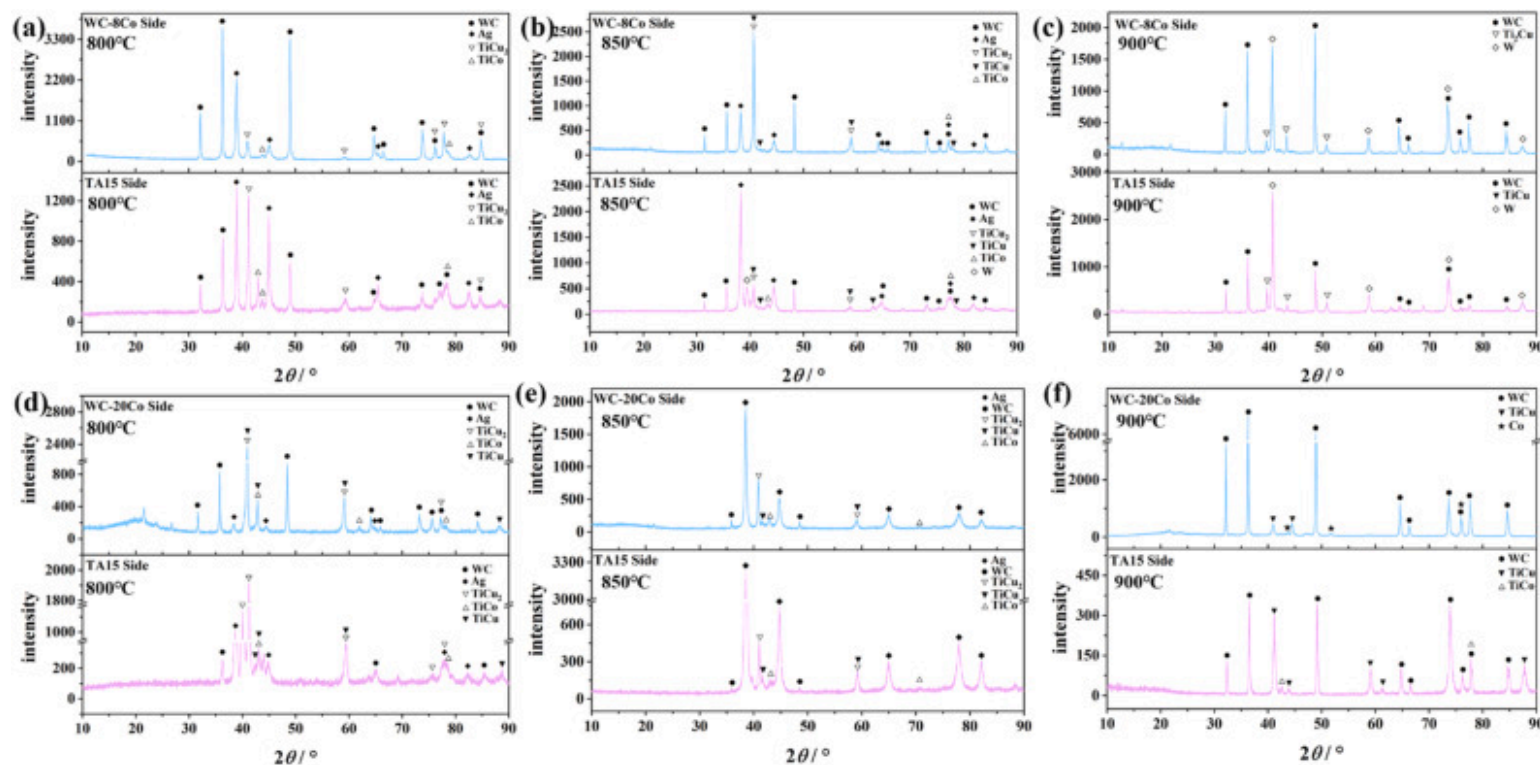
Fig. 13. Shear surface micromorphology and EDS analysis results of TA15/WC-xCo composite plate brazed at 850°C, (a) and (b) are respectively the TA15 side and the WC-8Co side of the TA15/WC-8Co composite plate, (c) and (d) are respectively the TA15 side and the WC-20Co side of the TA15/WC-20Co composite plate, (a₁, b₁, c₁, d₁) are respectively the magnified views of (a, b, c, d).



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Fig. 14. Shear surface micromorphology and EDS analysis results of TA15/WC-xCo composite plate brazed at 900°C, (a) and (b) are respectively the TA15 side and the WC-8Co side of the TA15/WC-8Co composite plate, (c) and (d) are respectively the TA15 side and the WC-20Co side of the TA15/WC-20Co composite plate, (a1, b1, c1, d1) are respectively the magnified views of (a, b, c, d).



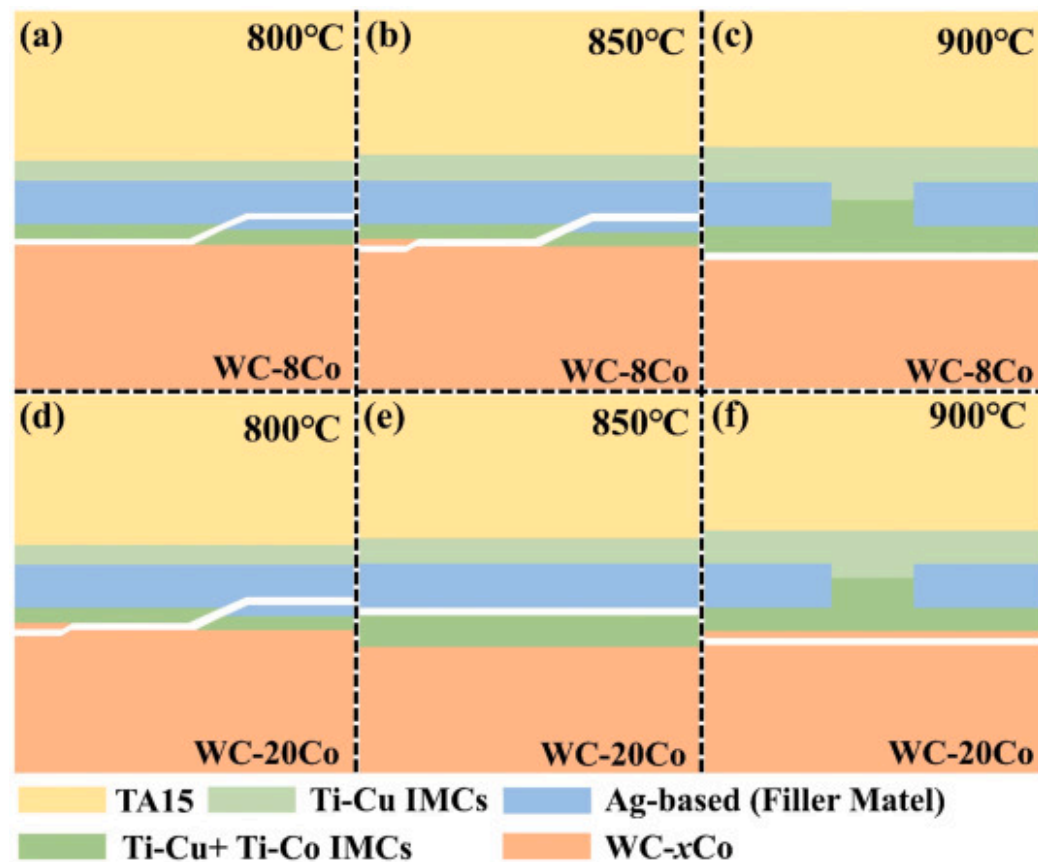
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Fig. 15. The diffraction peak of TA15/WC-xCo composite plate shear surface was analyzed based on X-ray diffraction at different brazing temperatures, (a) and (d) are TA15/WC-xCo composite plates brazed at 800°C, (b) and (e) are TA15/WC-xCo composite plates brazed at 850°C, (c) and (f) are TA15/WC-xCo composite plates brazed at 900°C.

At 800°C, the shear surfaces on both sides of the TA15/WC-8Co composite plate consist mainly of two parts: Ag-based (filler metal) and mixed areas of Ti-Cu, Ti-Co IMCs, and WC. The WC, in small and medium-sized blocks, is discretely distributed between Ti-Cu and Ti-Co IMCs in the mixed region, as shown in Fig. 12(a and b). This indicates that during the shearing process, the WC particles and Co binder near the interface are separated from the WC-8Co substrate. Additionally, the morphology of the Ag-based (filler metal) area on both shear surfaces is characterized by shear stripe bands parallel to the shear direction. Combined with the XRD results, the shear surfaces on both sides are mainly composed of WC, Ag, TiCu₂, and TiCo phases.

Therefore, the shear failure surface of the TA15/WC-8Co composite plate is located at the TiCu_2 (main) and TiCo mixed layer/WC-8Co interface, as well as the Ag-based (filler metal) layer. The failure schematic diagram is shown in Fig. 16(a). For the TA15/WC-20Co composite plate, the shear surfaces on both sides consist mainly of three areas: the WC-20Co, Ti-Cu, Ti-Co IMCs mixing zone, and Ag-based (filler metal). The Ag-based (filler metal) area also exhibits shear fringe bands parallel to the shear direction, but the mixed areas of Ti-Cu and Ti-Co IMCs show a continuous lamellar distribution. Additionally, there are distinct WC-20Co layer morphology features at the edge of the shear specimen, as shown in Fig. 12(d). Combined with the XRD diffraction results, the shear surfaces on both sides are mainly composed of WC, Ag, TiCu_2 , TiCu, and TiCo phases. This indicates that there are three shear separation surfaces when the TA15/WC-20Co composite plate is sheared. The first separation surface is the TiCu_2 , TiCo IMCs hybrid layer/WC-20Co interface, the second separation surface is the TiCu_2 , TiCo hybrid layer/ TiCu_2 layer interface, and the third separation surface is the Ag-based (filler metal) layer itself. The failure schematic diagram is shown in Fig. 16(d).



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Fig. 16. Schematic diagram of shear failure surfaces of TA15/WC-xCo composite plates brazed under different conditions, (a) TA15/WC-8Co composite plate brazed at 800°C, (b) TA15/WC-8Co composite plate brazed at 850°C, (c) TA15/WC-8Co composite plate brazed at 900°C, (d) TA15/WC-20Co composite plate brazed at 800°C, (e) TA15/WC-20Co composite plate brazed at 850°C, (f) TA15/WC-20Co composite plate brazed at 900°C.

At 850°C, the shear fracture surfaces on both sides of the TA15/WC-8Co composite plate consist of the Ag-based (filler metal), WC particles, and mixed regions of Ti-Cu/Ti-Co intermetallic compounds (IMCs). Notably, the WC layer exhibits a discontinuous blocky distribution between the Ag-based (filler metal) and the Ti-Cu/Ti-Co IMCs mixture, as illustrated in Fig. 13(a). XRD analysis confirms the predominant presence of WC, Ag, TiCu₂, TiCu, and TiCo phases on both fracture surfaces. Therefore, the TA15/WC-8Co composite plate has three shear separation surfaces. The first separation surface is in the WC-8Co matrix near the interface, the second is in the Ti-Cu and Ti-Co IMC layers, and the third is in the Ag-based (filler metal) layer itself. The failure schematic diagram is shown in Fig. 16(b). For the TA15/WC-20Co composite plate, the shear surface is mainly composed of Ag-based (filler metal) and Ti-Cu, Ti-Co IMCs layers. Based on the XRD results, it was found that the shear surfaces on both sides mainly contained Ag, TiCu₂, TiCu, and TiCo phases, along with a small amount of WC phase. The shear failure interface of the TA15/WC-20Co composite plate is the Ag-based (filler metal)/Ti-Cu, Ti-Co IMCs layer interface and the Ag-based (filler metal) itself. The failure schematic diagram is shown in Fig. 16(e). The results show that the TA15/WC-20Co composite plate prepared at 850°C has good interfacial bonding properties.

At 900°C, the shear surfaces on both sides of the TA15/WC-8Co composite plate are composed of block-shaped Ti-Co and Ti-Cu IMC laminates, randomly mixed with WC, with WC being the predominant phase, as shown in Fig. 14(a) and (b). It is worth noting that the Ti-Co and Ti-Cu IMCs present on the shear surface are primarily distributed in blocky forms, randomly arranged between the WC. Combined with the XRD results, it was found that the shear surfaces on both sides were mainly composed of WC, Ag, and Ti₂Cu phases. This indicates that the shear separation surface is located at the WC-8Co/(Ti-Co, Ti-Cu) IMC interface. The failure schematic diagram is shown in Fig. 16(c). However, for the TA15/WC-20Co composite plate, the shear surfaces on both sides are almost entirely composed of WC, indicating that the shear separation surface is located at the WC-20Co substrate near the interface. This also corresponds to the long cracks parallel to the interface formed in the WC-20Co near the interface (Fig. 16(f)), as observed earlier.

In conclusion, at relatively low temperatures (800°C–850°C), TA15/WC-xCo composite plates all exhibit good interface bonding performance. Meanwhile, as the Co content increases, the interface bonding performance can be significantly improved. One reason is that the Co and Ag elements undergo significant "replacement," ensuring the structural and strength integrity of WC-20Co near the interface, which enhances the interface bonding performance. Another reason is that a more stable diffusion layer is formed, which further contributes to improving the interface bonding performance. However, as the brazing temperature increases (900°C), the interfacial bonding performance of TA15/WC-xCo composite plates significantly decreases, and the bonding performance worsens with the increase in Co content (WC-20Co). One reason for this is the evaporation of Ag-based (filler metal), which leads to its discontinuous distribution at the interface, while a relatively thick IMC layer is simultaneously formed at the interface. During cooling shrinkage, a large number of crack defects form between the IMC layers. For composite plates with high Co content, due to the diffusion of Co elements and the absence of Ag element "replacement," a large quantity of voids and defects appear in the WC-20Co at the interface. Under the influence of thermal stress during cooling shrinkage, WC-20Co generates long cracks parallel to the interface, which severely affects the interface bonding performance. Through this study, TA15/WC-xCo composite plates were prepared by brazing Ag-Cu-Ti (filler metal) at a relatively low temperature (800°C–850°C) and using cemented carbide with high Co content, which can achieve relatively good interface connection [31].

4. Conclusion

In this work, TA15/WC-xCo composite plates were fabricated by brazing using Ag-Cu-Ti (filler metal). The microstructure was analyzed using SEM, EBSD, EDS, and XRD. The interfacial bonding performance of the composite materials was evaluated through compression shear tests. The effects of different brazing temperatures (800°C, 850°C, 900°C) and Co content (8% and 20%) on the interfacial microstructure and shear properties of TA15/WC-xCo were investigated. Additionally, the magnitude and distribution of the interface stress of the composite plate were studied and analyzed using the finite element simulation method. Following main conclusions are drawn.

- (1) As the brazing temperature increases (800°C–900°C), the grains of TA15 titanium alloy grow, but the local orientation differences between the grains decrease. Meanwhile, its UTS and elongation also decrease with the rise in temperature as results of the grain coarsening.
- (2) For cemented carbide with low Co content (WC-8Co), as the brazing temperature rises, the interfacial shear strength first increases and then decreases. At 850°C, the shear strength can reach 183.1 MPa. For cemented

carbide with high Co content (WC-20Co), the interfacial shear strength decreases as the brazing temperature rises. The shear strength at 800°C can reach 238.2MPa, and there is not much difference in shear strength between 800°C and 850°C.

- (3) At a relatively low brazing temperature (800°C–850°C), thin layer of Ti-Cu and Ti-Co IMCs will form, which contribute to form a good interface bonding. At a relatively high temperature (900°C), continuous and overly thick brittle Ti-Cu and Ti-Co IMCs will form, severely affecting the shear properties of the interface.
- (4) Based on the finite element simulation analysis, thermal stress of the Mises, S11, and S33 stresses on the plate gradually increase from the center to the edge. Meanwhile, interlayer cracks will occur at the interface of the composite plate during the cooling process due to the influence of shrinkage stress.

Overall, at a relatively low brazing temperature (800°C–850°C), Ag-Cu-Ti (filler metal) can reliably connect TA15 and WC-xCo.

CRedit authorship contribution statement

Yuhe Tian: Writing – review & editing, Writing – original draft, Validation, Investigation, Data curation. **Shunjie Jia:** Validation, Investigation, Data curation. **Yuqi Liu:** Validation, Investigation, Data curation. **Yang Feng:** Validation, Investigation, Data curation. **Weijun He:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Jun Peng:** Writing – review & editing, Supervision, Funding acquisition. **Bin Jiang:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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